

Your Code

```
1 import java.util.*;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=Integer.parseInt(sc.nextLine().trim());
6         Employee[] emp =new Employee[n];
7         for(int i=0;i<n;i++){
8             String type=sc.nextLine().trim();
9             while(type.isEmpty())
10                 type=sc.nextLine().trim();
11             int id=Integer.parseInt(sc.nextLine().trim());
12             String name=sc.nextLine().trim();
13             String dept=sc.nextLine().trim();
14             double basicSalary=Double.parseDouble(sc.nextLine().trim());
15             int value1=Integer.parseInt(sc.nextLine().trim());
16             double value2=Double.parseDouble(sc.nextLine().trim());
17             if(type.equalsIgnoreCase("TeachingFaculty")){
18                 emp[i]=new TeachingFaculty(id,name,dept,basicSalary,value1,value2);
19             }else if(type.equalsIgnoreCase("TechnicalStaff")){
20                 emp[i]=new TechnicalStaff(id,name,dept,basicSalary,value1,value2);
21             }else if(type.equalsIgnoreCase("VisitingFaculty")){
22                 emp[i]=new VisitingFaculty(id,name,dept,basicSalary,value1,value2);
23             }
24         }
25     }
26
27     String temp=sc.nextLine().trim();
28
29     while(temp.isEmpty())
30         temp=sc.nextLine().trim();
31     int sid;
32     if(temp.equalsIgnoreCase("Search Employee ID:"))
33         sid=Integer.parseInt(sc.nextLine().trim());
34     else
35         sid=Integer.parseInt(temp);
36     System.out.println("Employee Details");
37     System.out.println();
38     double total=0;
39     Employee highest=emp[0];
40     for(Employee e:emp){
41         System.out.println("Employee ID : "+e.getEmployeeId());
42         System.out.println("Name : "+e.getEmployeeName());
43         System.out.println("Department : "+e.getDepartment());
44         System.out.println("Monthly Salary : "+e.calculateSalary());
45         total+=e.calculateSalary();
46         if(e.calculateSalary()>highest.calculateSalary())
47             highest=e;
48     }
49     System.out.println("Highest Paid Employee");
50     System.out.println("Employee ID : "+highest.getEmployeeId());
51     System.out.println("Salary : "+highest.calculateSalary());
52     System.out.println();
53     System.out.println("Average Salary");
54     System.out.printf("%.2f\n",total/n);
55     System.out.println();
56
57     boolean f=false;
58
59
60     for(Employee e:emp){
61         if(e.getEmployeeId()==sid){
62             System.out.println("Employee Found");
63             System.out.println("Employee ID : "+e.getEmployeeId());
64             System.out.println("Name : "+e.getEmployeeName());
65             System.out.println("Salary : "+e.calculateSalary());
66             f=true;
67             break;
68         }
69     }
70
71     if(!f){
72         System.out.println("Employee Not Found");
73     }
74     sc.close();
75 }
76
77 class Employee{
78     private int employeeId;
79     private String employeeName;
80     private String department;
81     private double basicSalary;
82     Employee(int employeeId,String employeeName,String department,double basicSalary){
83         this.employeeId=employeeId;
84         this.employeeName=employeeName;
85         this.department=department;
86         this.basicSalary=basicSalary;
87     }
88
89     public int getEmployeeId(){
90         return employeeId;
91     }
92     public String getEmployeeName(){
93         return employeeName;
94     }
95     public String getDepartment(){
96         return department;
97     }
98 }
```

Input

```
3
TeachingFaculty
101
```

Output

```
Employee Details
Employee ID : 101
Name : Dr. Arun
Department : CSE
Monthly Salary : 60000.0
```

```

98     public double getBasicSalary(){
99         return basicSalary;
100     }
101     public double calculateSalary(){
102         return basicSalary;
103     }
104 }
105 class TeachingFaculty extends Employee{
106     private int teachingHours;
107     private double incentivePerHour;
108     public TeachingFaculty(int id,String name,String dept,double salary){
109         super(id,name,dept,salary);
110         teachingHours=h;
111         incentivePerHour=p;
112     }
113
114     public double calculateSalary(){
115         return getBasicSalary()+(teachingHours*incentivePerHour);
116     }
117 }
118 }
119 class TechnicalStaff extends Employee{
120     private int overtimeHours;
121     private double overtimeRate;
122     public TechnicalStaff(int id,String name,String dept,double salary){
123         super(id,name,dept,salary);
124         overtimeHours=h;
125         overtimeRate=r;
126     }
127
128     public double calculateSalary(){
129         return getBasicSalary()+(overtimeHours*overtimeRate);
130     }
131 }
132 class VisitingFaculty extends Employee{
133     private int lectureHandled;
134     private double paymentPerLecture;
135     public VisitingFaculty(int id,String name,String dept,double salary){
136         super(id,name,dept,salary);
137         lectureHandled=l;
138         paymentPerLecture=p;
139     }
140
141     public double calculateSalary(){
142         return lectureHandled*paymentPerLecture;
143     }
144 }
145 }

```

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Your Code

```

1 import java.util.*;
2 public class Main{
3
4     public static void main(String[] args){
5         Scanner sc=new Scanner(System.in);
6         int n=Integer.parseInt(sc.nextLine().trim());
7         Vehicle[] v=new Vehicle[n];
8         for (int i =0;i<n;i++){
9             String type=sc.nextLine().trim();
10            while(type.isEmpty())
11                type=sc.nextLine().trim();
12            String number=sc.nextLine().trim();
13            String owner=sc.nextLine().trim();
14            double amount=Double.parseDouble(sc.nextLine().trim());
15            if(type.equalsIgnoreCase("Car"))
16                v[i]=new Car (number,owner,amount);
17            else if(type.equalsIgnoreCase("Bike"))
18                v[i]=new Bike(number,owner,amount);
19            else if(type.equalsIgnoreCase("Truck"))
20                v[i]=new Truck(number,owner,amount);
21
22        }
23        String temp=sc.nextLine().trim();
24        while(temp.isEmpty()){
25            temp=sc.nextLine().trim();
26        }
27        String search;
28        if(temp.equalsIgnoreCase("Search vehicle:"))
29            search=sc.nextLine().trim();
30        else
31            search=temp;
32        Vehicle highest=v[0];
33        for(Vehicle x:v){
34            System.out.println("Vehicle : "+x.getVehicleNumber());
35            System.out.println("Claim Amount : "+x.calculateClaimAmount());
36            System.out.println();
37            if(x.calculateClaimAmount()>highest.calculateClaimAmount())
38                highest=x;
39        }
40
41        System.out.println("Highest Claim Vehicle");
42        System.out.println(highest.getVehicleNumber());
43        System.out.println(highest.calculateClaimAmount());
44        System.out.println();
45        boolean found=false;
46        for(Vehicle x:v){
47            if(x.getVehicleNumber().equals(search)){
48                found=true;
49                System.out.println("Vehicle Found");
50                System.out.println("Owner : "+x.getOwnerName());
51                System.out.println("Claim Amount : "+x.calculateClaimAmount());
52                break;
53            }
54        }
55        if(!found){
56            System.out.println("Vehicle Not Found");
57        }
58        sc.close();
59    }
60 }
61
62 class Vehicle{
63     private String vehicleNumber;
64     private String ownerName;
65     private double insuredAmount;
66     public Vehicle(String vehicleNumber,String ownerName,double insuredAmount){
67         this.vehicleNumber=vehicleNumber;
68         this.ownerName=ownerName;
69         this.insuredAmount=insuredAmount;
70     }
71     public String getVehicleNumber(){
72         return vehicleNumber;
73     }
74     public String getOwnerName(){
75         return ownerName;
76     }
77
78     public double getInsuredAmount(){
79         return insuredAmount;
80     }
81     public double calculateClaimAmount(){
82         return insuredAmount;
83     }
84 }

```

Input

```

3
Car
TN10AB1234

```

Output

```

Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN09XY5678
Claim Amount : 60000.0

```

```
84
85 }
86 class Car extends Vehicle{
87     public Car(String v,String o,double a){
88         super(v,o,a);
89     }
90 }
91 public double calculateClaimAmount(){
92     return getInsuredAmount()*0.80;
93 }
94 }
95 class Bike extends Vehicle{
96     public Bike(String v,String o,double a){
97         super(v,o,a);
98     }
99     public double calculateClaimAmount(){
100         return getInsuredAmount()*0.60;
101     }
102 }
103
104 }
```

```
105 class Truck extends Vehicle{
106     public Truck(String v,String o,double a){
107         super(v,o,a);
108     }
109 }
110 public double calculateClaimAmount(){
111     return getInsuredAmount()*0.90;
112 }
113 }
```

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